

Power Distribution System Planning with Reliability Modeling and Optimization

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Abstract - A new approach for the systemized optimization of power distribution systems is presented in this paper. Distribution system reliability is modelled in the optimization objective function via outage costs and costs of switching devices, along with the nonlinear costs of investment, maintenance and energy losses of both the substations and the feeders. The optimization model established is multi-stage, mixed-integer and nonlinear, which is solved by a network-flow programming algorithm. A multi-stage interlacing strategy and a nonlinearity iteration method are also designed. Supported by an extensive database, the planning software tool has been applied to optimize the power distribution system of a developing city.

Key Words - power distribution system planning, reliability modeling and optimization, multi-stage decisions, nonlinear costs, network-flow programming, database

1. INTRODUCTION

The costs and reliability of power distribution systems are beginning to receive as much attention as those of power generation and transmission systems. Modern planning of these large-scale and complicated systems relies heavily on computers and mathematical optimization tools [12]. Especially in developing countries where the growing power demand provides needs for substantial expansion of power distribution systems, the long term benefit brought by using computerized optimization tools are tremendous, as compared to widely practised *ad hoc* methods involving only local computation.

The goal of modern power distribution system planning is to satisfy the growing and changing system load demand during the planning period and within operational constraints, economically, reliably and safely, by making optimized decisions on the following: voltage levels of the distribution network; locations, sizes, servicing areas, loads and building or expanding schedules of the substations; routes, conductor types, loads and building schedules of the sub-transmission lines and feeders; other important issues such as the types and locations of switching devices, load voltage levels, and network and load reliability levels, etc. The optimization problem is usually very complicated, considering the scale of the system and the many inter-related factors.

Many optimization models and solution methods have been proposed. They can be categorized in several ways [4,12]: by treatment of costs, as linear planning, mixed-integer planning (including 0-1 planning), nonlinear planning and network-flow planning; by treatment of planning periods, as static planning, pseudo-dynamic planning and dynamic planning; by treatment of planning durations, as long-term planning, mid-term planning and short-term planning.

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The investment and maintenance costs of the substations and the feeders are fixed costs (zero-order), while their operational costs are variable costs squarely depending on their loads (second-order) [3], making the objective function inherently nonlinear. [1] linearized both the zero-order term and the second-order term, and employed the tableau method used in transportation problems instead of the usual simplex method in linear programming. [2] set up a mixed-integer optimization model, linearizing only the second-order term and small zero-order terms while large zero-order terms remain, and it was solved by the branch-and-bound method. [3] proposed a nonlinear mixed-integer model, keeping the cost nonlinearity, and treated the discrete decision variables as continuous ones to utilize linear programming method, and then discretized those decision variables.

Static planning [1,2,3] considered only a single planning period, while dynamic planning minimizes overall present value of several planning periods, achieving a more precise and optimal modeling. Every stage in the dynamic planning interacts with each other, thus making dynamic planning a much more difficult task. Methods in the literature dealing with dynamic planning usually make approximations, since the orders of distribution systems are very large and the number of states is unpredictable, while precise load forecasting is unavailable. [4] used dynamic programming method to search for the optimal or suboptimal combinations of decisions from many possible, instead of optimal, solutions, obtained by independent static planning of every stages. [5] used the branch-and-bound algorithm on an integer optimization model to select and limit the possible serials of decisions. [6] employed successive concatenated optimization technique to minimize the overall objective function. [7] used a decomposition and coordination algorithm based on a branch exchange method.

A major difficulty that exists in modern power distribution system planning is the conflict between the precision of the model and the computation efficiency of the solution method for the model, which is even worsen by the availability and accuracy of the data required by the model. A new approach for the systemized optimization of large-scale power distribution systems is presented in this paper. Techniques are designed to effectively compromise the difficulties brought by the nonlinearity and dynamical nature of the large-scale problem. These techniques are incorporated into the network-flow programming algorithm. The network-flow algorithm had previously been used in distribution planning, but as a static linear programming method [8].

The need to seriously consider reliability in terms of capital costs in distribution system planning is being recognized, as emphasized recently by an IEEE task force on distribution reliability modeling and applications [14]. Unfortunately, few papers had extensively tackled this problem [9,10]. In this paper, reliability modeling is incorporated into the overall objective function via outage costs and costs of switching devices, along with the present value of the costs of investment, maintenance and energy losses for both the substations and the feeders.

Constraints for the objective function established in this paper include Kirchhoff's current law, power capacity limits, voltage drop limits, radiality constraint and reliability indices [3,5,6,7]. The established multi-stage mixed-integer nonlinear model is solved by the network-flow programming algorithm with a multi-stage interlacing strategy and a nonlinearity iterating function. Supported by an extensive database, the planning tool had been successfully used in the optimization of the power distribution system of a developing city, and part of the earlier results are given in this paper.

II. RELIABILITY MODELING AND OPTIMIZATION

1) Reliability optimization of power distribution system

Quantitative reliability measurement is being recognized as necessary and is becoming feasible in the planning of power distribution systems [14]. The improvement in system reliability level, or the decreasing of outage costs, usually demands an increase in investment costs. The goal of the reliability optimization is to search for the minimal equilibrium, as indicated in Fig. 1.

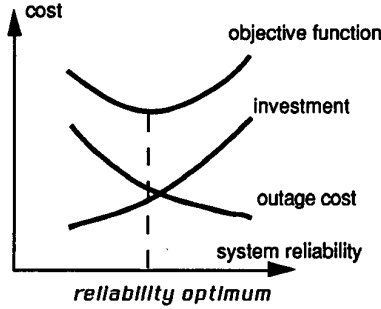


Fig. 1 Reliability Optimization

The reliability optimization is incorporated into the overall optimization in this paper. Considering static planning first in this section, the objective function of the distribution system optimization can be written as

$$\text{Min } Z = \sum_{i=1}^{N_s+N_f} (F_i + C_i + R_i + \psi_i S_i) \quad (1)$$

where Z : total present value of system cost
 N_s, N_f : total number of substations and feeders, respectively
 F_i : fixed cost of substation or feeder i
 C_i : variable cost of substation or feeder i
 R_i : outage cost of substation or feeder i
 ψ_i : decision on switching devices in substation or feeder i
 S_i : investment cost of switching device in substation or feeder i

A circuit breaker is considered as a switch, and ψ_i takes the values of 0 and 1 for no switch and with switch, respectively. The separation switches are not considered; otherwise, the switch operation time affects the computation of outage costs and complicates the reliability optimization.

2) The computation of outage costs

The outage costs are modelled as a function of outage frequency, outage duration, averaged outage power and energy. For each branch, either a substation or a feeder, its outage cost is related not only to its own outages but also to outages in other branches. Therefore, the factors affecting outage costs include: network configuration, network element reliabilities, the locations and reliability of switches, lost load and its capital value.

When there is no switch in between, the outage cost of any branch i due to the outage of branch j can be written as (note the special case $i = j$, which indicates the outage cost of the branch as incurred by its own outage)

$$R_i = a \lambda_j P_i + b U_j P_i \quad (2)$$

where a : outage cost per MW per outage
 b : outage cost per MW per hour
 λ_j : average annual outage frequency of branch j
 U_j : average annual outage duration of branch j
 P_i : average lost load of branch i due to outage, in MW

Since outages in branches affect each other, further complicated by the switches, the computation of the outage costs warrants an integrated and systematic approach.

a. Single feeder with single layer branches

The simplest case is shown in Fig. 2. The outage cost without or with switch B is computed by (3) and (4), respectively,

$$R = \sum_{j=1}^k (a \lambda_j + b U_j) P_1 \quad (3)$$

$$R = \sum_{j=1}^{k-1} (a \lambda_j + b U_j) P_1 + (a \lambda_B + b U_B) P_1 + (a \lambda_k + b U_k) P_k + S_B \quad (4)$$

where S_B : cost of switch B

λ_B : average annual outage frequency of switch B

U_B : average annual outage duration of switch B

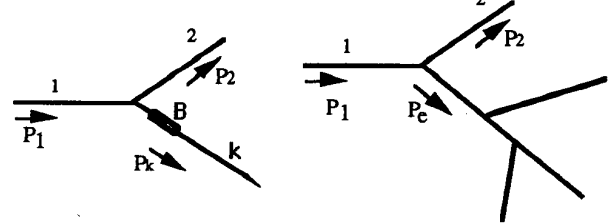


Fig. 2 (a) Branches with switch B (b) Multilayer branches

b. Radial multi-layer branches with at most a single switch

For radial multi-layer branches with a single switch, the branch with a switch should be normalized to one without a switch:

$$R = \sum_{j=1}^{k-1} (a \lambda_j + b U_j) P_1 + (a \lambda_e + b U_e) P_1 \quad (5)$$

in which

$$\lambda_e = \lambda_B + \lambda_k \frac{P_k}{P_1} \quad (6)$$

$$U_e = U_B + U_k \frac{P_k}{P_1} + \frac{S_B}{b P_1} \quad (7)$$

After the switches, if any, are normalized, multi-layer branches, as shown in Fig. 2(b), should then be combined by simply add the outage rates and outage durations of all the branches. The generalized equation for outage cost is then

$$R = a \lambda_e P_1 + b U_e P_1 \quad (8)$$

or

$$R = \beta P_1 \quad (9)$$

where $\beta = a \lambda_e + b U_e$ is termed outage cost coefficient.

c. Multi-layer branches with more than one switch

The situation where there are more than one switch in the radial branches is more complicated, since the number and locations of switches significantly affect outage costs. Assume in Fig. 3 that there is a switch in branch m , the outage cost of branch k without or with switch B in branch k is computed by (10) and (11), respectively,

$$R_k = (a \lambda_k + b U_k) P_m \quad (10)$$

$$R_k = (a \lambda_B + b U_B) P_m + (a \lambda_k + b U_k) P_k + S_B \quad (11)$$

The solution method for the overall outage costs of related branches is given in the next section.

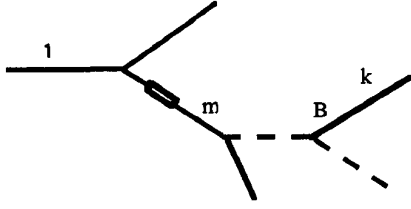


Fig. 3 Multilayer branches with switches

III. PLANNING MODEL AND INTEGRATED SOLUTION

1) Objective function

Dynamical objective function can be written as

$$\text{Min } Z = \sum_{j=1}^Y \left(\sum_{i=1}^N (F_{ij} + C_{ij} + R_{ij} + \psi_{ij} S_{ij}) \right) \quad (12)$$

where Y: total planning stage

N: total number of branches ($N = N_S + N_F$)

$F_{ij}, C_{ij}, R_{ij}, \psi_{ij}, S_{ij}$: defined in (1) at planning stage j

This nonlinear, mix-integer optimization model can be simulated by the following linearized model,

$$\text{Min } Z = \sum_{j=1}^Y \left(\sum_{i=1}^N (\alpha_{ij} P_{ij} + \beta_{ij} P_{ij} + \psi_{ij} S_{ij}) \right) \quad (13)$$

where α_{ij} is termed dynamic cost coefficient. The first term in (13) stands for the submodel of the investment, maintenance and operation costs of the substations and feeders. Two interlaced steps of iteration computations are operated on α_{ij} . One step is the multi-stage interlacing, in which α_{ij} of the same element i but of different planning stage j are adjusted and coordinated throughout the planning duration. The other step is the nonlinearity iterating, in which α_{ij} of the same planning stage j are iteratively solved to better approximate the fixed costs and nonlinear operation costs by a linearized coefficient.

To explain the techniques used to solve the overall optimization problem, reliability optimization submodel is separated from (13) in the following developments. However, it should be noted that the iterations of α_{ij} and β_{ij} are inter-related in the network-flow optimization process.

2) Solution method for reliability optimization

The outage costs in different planning stages for a same element i are independent of each other, thus no adjustment is needed for β_{ij} between different stages. For this reason, β_{ij} is simplified in the following as β_i . However, in a certain stage j, β_{ij} depends on P_{ij} and ψ_{ij} , and on each other.

Network-flow programming process [11] facilitates the iteration of β_i . The optimal augmented chain is sought and the flow amount is increased in the chain; this process goes on until the given flow is met or the maximum flow is reached. To solve for β_i in (13), $\mu_{ij} (= \alpha_{ij} + \beta_i)$ is selected as the arc length and the shortest path technique can be used to find the optimal augmented chain. Then using the flow, the outage costs and switch information on the chain, and the outage cost coefficients of each branch in the chain, the switch decision is optimized and β_i of the associated branches are adjusted.

Assume that all the feeders directly from the substations have switches and their outage rates are neglected. As a result, there is no need to adjust β_i and optimize ψ_i for substation branches. Except

for substation branches, the outage cost for a chain can be obtained by summing up outage costs of every arcs in the chain, while the arcs with a switch are normalized to ones without a switch, as shown in (3) and (5). Outage cost is also affected by the existence of connected arcs that are not in the chain; however, from (3) and (4), it can be seen that the effect is the same for outage damage in the chain, thus it can be ignored in the process of searching for the shortest path.

If using β_i as the arc length to search for the shortest path, the minimum of $\sum_{i=1}^s \beta_i$ is obtained, such that the following minimum outage cost is obtained

$$R = \min \sum_{i=1}^s \beta_i (P_1^{(k)} + \min \{ \Delta_i \}) \quad (14)$$

where s: the number of arcs in the shortest path

$P_1^{(k)}$: the flow on the arc closest to the substation branch; k denotes the kth iteration

Each time the optimal augmented chain is found, the flow on the chain is increased the amount that is allowable to the arc with the minimal flow. Then from the furthestmost arc from the substation branch closing on to the closest one, the switch decisions are optimized one by one by comparing R with the following outage costs:

a. switch in arc j, and there is no switch in the arcs between j and the substation branch:

$$R = \sum_{i=1}^{j-1} \beta_i P_1 + (a\lambda_B + bU_B) P_1 + \sum_{i=j}^s \beta_i P_j + S_B \quad (15)$$

b. switch in arc j, and there is a switch in an arc m between j and the substation branch:

$$R = \sum_{i=1}^{m-1} \beta_i P_1 + (a\lambda_{Bm} + bU_{Bm}) P_1 + \sum_{i=m}^{j-1} \beta_i P_m + \sum_{i=j}^s \beta_i P_j + (a\lambda_{Bj} + bU_{Bj}) P_1 + S_B \quad (16)$$

After the switch decisions for the arcs in the optimal augmented chain have been made (though temporally) and new flow has been computed, outage cost coefficient β_i for each arc are adjusted beginning from the furthestmost arc. Arcs with switches should be normalized. For example, (15) is normalized to be

$$R = \sum_{i=1}^{j-1} \beta_i P_1 + a\lambda_{je} P_1 + bU_{je} P_1 + a \sum_{i=j+1}^s \lambda_{ie} P_1 + b \sum_{i=j+1}^s U_{ie} P_1 \quad (17)$$

in which

$$\lambda_{je} = \lambda_B + \lambda_j \frac{P_j}{P_1} \quad (18)$$

$$U_{je} = U_B + U_j \frac{P_j}{P_1} + \frac{S_B}{bP_1} \quad (19)$$

$$\lambda_{ie} = \lambda_i \frac{P_i}{P_1} \quad (i=j+1, \dots, s) \quad (20)$$

$$U_{ie} = U_i \frac{P_i}{P_1} \quad (i=j+1, \dots, s) \quad (21)$$

After a new series of β_i has been obtained, the shortest path method is again used to find the optimal augmented chain and the flow is increased in the chain, then the switch decisions are optimized and the switches are normalized, to obtained a newer series of β_i . This process is continued until all the loads are

satisfied.

3) Nonlinear iterating function

Excluding reliability optimization submodel from (12), then the nonlinear and mix-integer objective function can be written as

$$\text{Min } Z = \sum_{i=1}^Y \left(\sum_{j=1}^N (F_{ij} + e_{ij} P_{ij}^2) \right) \quad (22)$$

where e_{ij} : annual operation cost coefficient of element i
 F_{ij}, P_{ij} : defined in (12) and (13)

As explained earlier, (13) is iteratively linearized by α_{ij} . From the $(k-1)$ th iteration,

$$Z_{ij}^{(k-1)} = \alpha_{ij}^{(k-1)} P_{ij}^{(k-1)} \quad (23)$$

$$\text{then } \alpha_{ij}^{(k)} = \frac{F_{ij} + e_{ij} P_{ij}^{(k-1)2}}{P_{ij}^{(k-1)}} \quad (24)$$

If $P_{ij}^{(k-1)}=0$, $\alpha_{ij}^{(k)} = \alpha_{ij}^{(k-1)}$. The initial values of α_{ij} can be obtained by using the rated power P_{ijN} in (24). After several iterations, (23) would better approximate, than the simple one-step linearization method, the respective terms in (22) at the operation points of each branch. Criterion $|Z^{(k)} - Z^{(k-1)}| < \epsilon$ is provided for the iterations, which are incorporated in the multi-stage interlacing strategy, as explained later.

In addition to bringing a better approximation to the real cost than the simple linearization method, the rationality for the nonlinearity iteration method is enhanced by its physical implications. Differentiating (24), it yields

$$\frac{d\alpha_{ij}}{dP_{ij}} = \frac{e_{ij}P_{ij}^2 - F_{ij}}{P_{ij}^2} \quad (25)$$

If $P_{ij}^2 > \frac{F_{ij}}{e_{ij}}$, the function $\alpha_{ij}(P_{ij})$ is strictly increasing in $[\frac{F_{ij}}{e_{ij}}, P_{ij\max}]$, which means that when the flow on an arc is greater than $\sqrt{\frac{F_{ij}}{e_{ij}}}$ and becoming too large, its cost coefficient increases automatically. This penalty process is mostly effective for feeders only, since their fixed cost F_{ij} is usually smaller than their operational cost $e_{ij}P_{ij}^2$.

If $P_{ij}^2 < \frac{F_{ij}}{e_{ij}}$, the function $\alpha_{ij}(P_{ij})$ is strictly decreasing in $[0, \frac{F_{ij}}{e_{ij}}]$. For feeders, when the power flow is smaller than $\sqrt{\frac{F_{ij}}{e_{ij}}}$ and becoming too small, its cost coefficient increases automatically, which means that feeders with a small power flow will be unfavored. For substations, since their fixed cost F_{ij} is usually greater than the operational cost $e_{ij}P_{ij}^2$, the higher their power flow, the lower their cost coefficients, which means that substations with greater capacities will be favored.

4) Multi-stage interlacing strategy

The problem of multi-stage optimization is decomposed into several single-stage optimization problems, and the subproblems are coordinated throughout the solution process, as shown in Fig. 4. The subproblems are related by the fixed costs of the elements, therefore the fixed costs F_{ij} are selected as the coordinators, for which combination optimizations and continuity examinations should be performed, as described in the following.

Multi-stage interlacing strategy uses the present value of the fixed cost F_{ij} as the coordinators and P_{ij} and F_{ij} as the inputs of the coordination process:

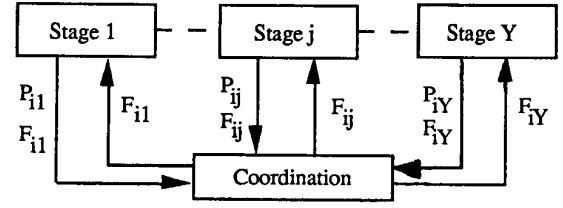


Fig. 4 Multistage Coordination

$$\text{If } P_{ij}^{(k-1)}=0, \text{ then } F_{ij}^{(k)} = \frac{Y - X_{ij}^{(k-1)}}{Y} F_{ij}^{(k-1)} \quad (26)$$

$$\text{If } P_{ij}^{(k-1)} \neq 0, \text{ then } F_{ij}^{(k)} = \frac{Y + W_{ij}^{(k-1)}}{Y} F_{ij}^{(k-1)} \quad (27)$$

where Y : the number of planning stages
 $X_{ij}^{(k-1)}$: number of non-zero $P_{im}^{(k-1)}$ ($m=1, \dots, l$; $m \neq j$)
 $W_{ij}^{(k-1)}$: number of zero $P_{im}^{(k-1)}$ ($m=1, \dots, l$; $m \neq j$)

In this way, the fixed cost of an arc in a planning stage is unfavored when this arc demands an element in this stage but does not demand an element in at least one other stage, and favored when this arc does not demand an element in this stage but demands an element in at least one other stage. The numbers $X_{ij}^{(k-1)}$ and $W_{ij}^{(k-1)}$ put weights on the favoring and unfavoring process.

Multi-stage coordination procedure is interlaced into the iterations of the dynamic cost coefficient α_{ij} . After each single-stage optimization by network-flow programming, α_{ij} is adjusted by the multistage coordinator F_{ij} and the arc flow P_{ij} , by the following formula:

$$\text{If } P_{ij}^{(k-1)}=0, \quad \alpha_{ij}^{(k)} = \frac{Y - X_{ij}^{(k-1)}}{Y} F_{ij}^{(k-1)} + e_{ij} P_{ijN}^2 \quad (28)$$

$$\text{If } P_{ij}^{(k-1)} \neq 0, \quad \alpha_{ij}^{(k)} = \frac{Y + W_{ij}^{(k-1)}}{Y} F_{ij}^{(k-1)} + e_{ij} P_{ij}^{(k-1)2} \quad (29)$$

5) The application of network-flow programming method

The distribution system can be seen as a network graph, with nodes and directed arcs, and the planning problem is a limited-capacity minimum-cost maximum-flow problem in the network-flow programming. Assume a source node and a sink node; assume artificial arcs between the source node and the substation nodes, with the maximum capacities and costs of the substations as their capacities and costs; assume artificial arcs between the sink node and all the nodes except substation nodes, with the load capacities as their capacities and with zero cost. When the overall flow into the sink node equals the sum of the loads, the network's minimum-cost flow pattern has been found.

The mathematical model for the network-flow programming can be written as

$$\text{Min } Z = \sum_{i=1}^N (\mu_i P_i + \psi_i S_i) \quad (30)$$

$$\text{s.t. } \sum_{i=M_{Ok}} P_i - \sum_{i=M_{Ik}} d_i P_i = 0 \quad (k=1, \dots, M; i \neq s, t) \quad (31)$$

$$\sum_{i=M_{Ot}} P_i - \sum_{i=M_{It}} d_i P_i = -P_t \quad (32)$$

$$0 \leq P_i \leq C_i \quad (i \in M) \quad (33)$$

where N: total number of arcs
M: number of original nodes
s: source node
t: sink node
 M_{Oj} : set of arcs leaving from node i
 M_{Ii} : set of arcs arriving at node i
 μ_i : linearized cost coefficient of arc i, $\mu_i = \alpha_i + \beta_i$
 d_i : increment of arc i, $d_i \leq 1$
 C_i : capacity of arc i
 P_t : overall flow into node t (i.e. total system load)

The essence of our modified network-flow programming is to select μ_i as the arc length, and use the shortest path technique to search for the shortest path π_k from node s to node t (i.e. the optimal cost augmented chain) with length $d\pi_k$. The flow on the chain is increased the amount that is allowable to the arc with the minimal flow, and the switch optimization and the adjustment of the outage cost coefficient β_i are performed, as explained previously.

The basic steps are listed in the following, in which Δ_i is the incremental flow, $\Delta_{\min}$ is the minimum incremental flow, μ_i^+ is the forward arc cost and μ_i^- is the backward arc cost:

- a. (1) If arc i is a forward arc in the s-t chain:
 - i) If $P_i < C_i$, then $\Delta_i = C_i - P_i$, $\mu_i^+ = \mu_i$, $\mu_i^- = \infty$ (if $P_i = 0$) or $-\mu_i$ (if $P_i > 0$)
 - ii) If $P_i = C_i$, then $\Delta_i = 0$, $\mu_i^+ = \infty$, $\mu_i^- = -\mu_i$
- (2) If arc i is a backward arc in the s-t chain:
 - i) If $P_i > 0$, then $\Delta_i = -P_i$, $\mu_i^+ = -\mu_i$
 - ii) If $P_i = 0$, then $\Delta_i = 0$, $\mu_i^- = \infty$
- b. $\Delta_{\min} = \min \{\Delta_i\}$ ($i = \pi_k$)
- If $P + \Delta_{\min} > P_t$, the maximum flow has been reached,
 $Z = Z + d\pi_k(P_t - P)$
else $P = P + \Delta_{\min}$, $Z = Z + d\pi_k \Delta_{\min}$
- c. For forward arc i, $P_i = P_i + \Delta_{\min}$; for backward arc i, $P_i = P_i - \Delta_{\min}$
- d. Optimization of ψ_i along π_k by (14) to (16)
- e. Adjustment of β_i according to flows and switch decisions of arc i and associated arcs further away from substations, by (3) to (9) and (17) to (21)

The above process is continued until the maximum flow P_t is reached. The initial value for P_i can be 0.

6) Additional Constraints

In addition to power balance constraint (or, Kirchhoff's current law) and power capacity limits as described previously, the voltage drop limits, radiality constraint and reliability constraint are also considered as constraints for the optimization of the objective function.

a. Voltage drop limits

Voltage drops have implicitly been constrained since the network power loss has been constrained along with the investment costs in the optimization process; however, it is still necessary to impose the voltage drop limits. The changing of conductor types is an effective method to meet these limits. Small conductors are selected first according to the power flows, and node voltages are calculated; in the case that any load voltage is lower than the specified limit, larger conductors replace the smaller ones starting from the substation output feeders one at a time, until the voltage limit is satisfied.

For the large loads with few alternative routes or long feeders, the voltage drop limits can have a significant impact on the optimization of substations and feeders. In this case, a new optimization process should be conducted after changing the conductor parameters.

b. Radiality constraint

In the operation of a distribution network, electricity is provided to loads radially to avoid electromagnetic loop, even though

tie-lines usually exist to enhance reliability. Radiality constraint that is required for normal operation is met if all the nodes except node s and node t are checked to have only one forward arc directed to each of them. Fig. 5(a) shows a typical violation of this constraint, in which P_L is the load.

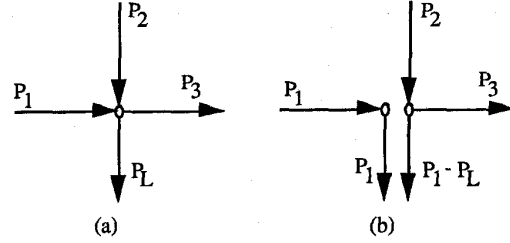


Fig. 5 Radiality constraint satisfaction

i) If $P_L > P_1$ or $P_L > P_2$: P_L can be divided into two parts, for example P_1 and $P_L - P_1$ in the case that $P_L > P_1$, as shown in Fig. 5(b).

ii) If $P_L < P_1$ and $P_L < P_2$: P_L can not be divided in this case. An arc with the smallest flow can be deleted in the loop, and another optimization process is conducted after voltage drop limit is checked and conductor types have been switched. An alternative method is to select the arc with switch and (or) smallest flow as open loop tie-line.

c. Reliability constraint

Reliability optimization aims at the reduction of the overall outage costs for the system. It is necessary to meet the reliability requirements of individual loads, by checking certain reliability indices for the loads, such as annual outage rates, average outage duration and annual outage duration. Though these specifications are not covered in this paper, the formula that is used to evaluate the overall reliability level of the optimized network is given below, for average supply availability indice (ASAI):

$$ASAI = \left(1 - \frac{\sum_{i=1}^M n_i U_i}{8760 \sum_{i=1}^M n_i}\right) \times 100\% \quad (34)$$

where M: total number of loads
 n_i : number of loads on node i
 U_i : annual outage duration of node i

IV. PLANNING TOOL: FLOW-CHART AND APPLICATION

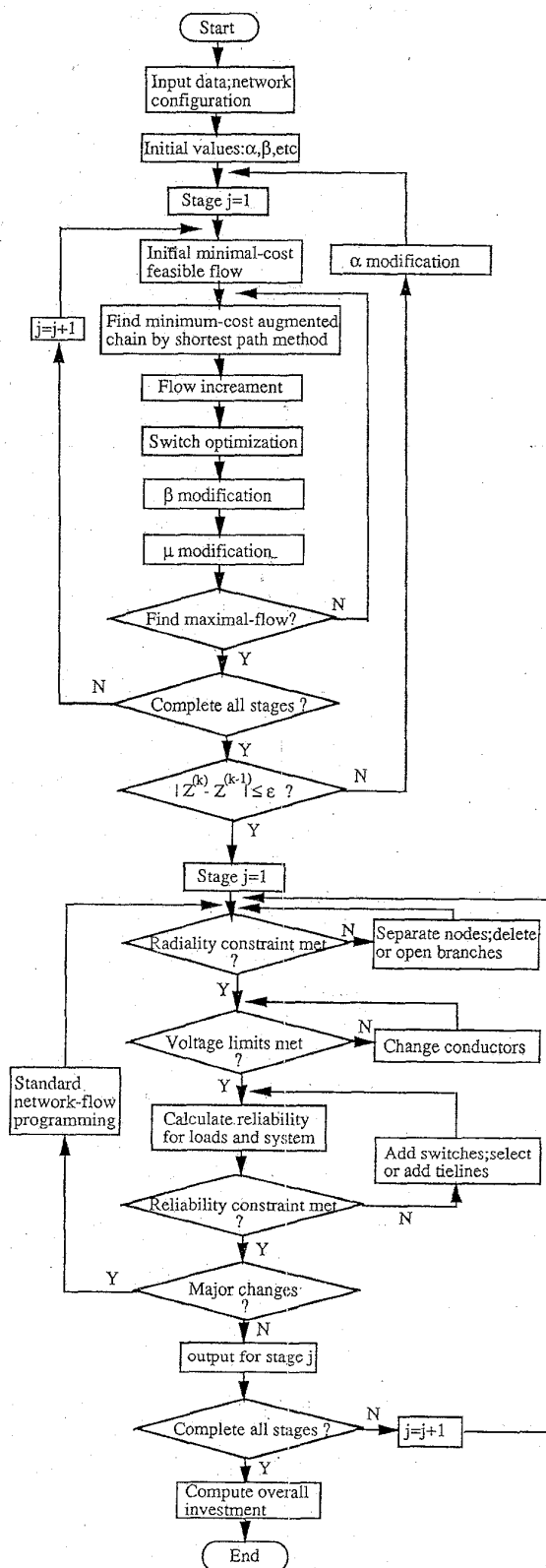
1) Flow-chart for computer programming

Based on the new approach, the flow-chart for computer programming is shown in the next page. Our programming language is Fortran 77, and the computer system is an IBM PC/AT 386. The major advantage of the network-flow programming method is its efficiency: in the optimization of the large-scale network described next, it took approximately 3 hours for a complete solution.

2) Application and database

The computer program based on the model and solution techniques have been applied in the optimization of the power distribution system of Shi-Jia-Zhuang City, the capital city of Hebei Province in China. The system excluding suburb area had 10 substations, feeders of 680km total, with maximum overall load of 300MW and average load density of 1171kw/km².

Several severe problems existed in the system, including: the construction of the system was lagging behind the increasing of loads, resulting in constant limitation and severe regulation of electricity usage; lacking of load transferring ability and operational flexibility, with very low reliability level (average supply availability at only 99.1%); there were some long feeders with heavy loads (one 10.5km feeder had a capacity of 22655kVA, while there was another



feeder with only 480kVA); voltage level was far from satisfactory; unreasonable configurations existed in the network, such that many loads were fed by other than a closer substation, and many feeders intercepted each others; high maintenance cost for the feeders (at approximately ¥380/km annually), due to the expired devices and messy conductor types.

The *ad hoc* planning procedure long exercised for the system is of short-term and operational nature, contributing to the free-wheeling of the system with unsatisfactory economical and operational performance. In addition to providing recommendations for system expanding decisions, the systemized and computerized optimization attempted to solve many of the existing problems. Part of the earlier testing result is summarized in next, which involves two stage short-term planning, each for 3 years.

A sufficient and accurate database is very important in validating the computerized "optimal" decision. A large portion of our work was devoted into carefully collecting the data. The inputs to the computer program were feeder configurations and possible routes, locations, present and maximum capacities of existing and new substations, forecasted loads, various costs, reliability data, etc.

The substation data is listed in Table I. The number of feeders including all branches was 627 and the number of nodes including 11 substation nodes was 398. Some of the important data are: cost of electricity 0.074¥/kW-hours; substation normal load is at 85% of its rated load; feeder annual outage rate 0.04/km, transformer annual outage rate 0.136; outage cost disregarding customer losses is 10¥/MW-times or ¥100/MW-hours.

Table I Substation Data

Substations	1	2	3	4	5	6	7	8	9	10	11
Capacity (MVA)	51.5	51.5	20	20	51.5	20	63	31.5	51.5	35	0
Expandable Capacity (MVA)	0	0	0	20 or 31.5	11.5	20 or 31.5	0	20 or 31.5	0	16.5 or 28	63
Investment (10 ⁴ ¥)	0	0	0	80 or 100	100	80 or 100	0	80 or 100	0	180 or 200	700
Peak Load of 1990 est. (MW)	36	36	12	34	27	18	26	15	28	26	/
Peak Load of 1995 est. (MW)	42	38	13	37	35	22	35	25	30	38	/

Two stage short-term optimization results for the substations are shown in Table II. Single stage optimization was also performed, with identical final year results as the two stage planning. System overall costs (present value) and reliability indice ASAI are listed in Table III.

Table II Substation Optimization Result (in MVA; '*' stands for expansion or new)

Substations	1	2	3	4	5	6	7	8	9	10	11
Multistage Planning Stage 1	51.5	51.5	20	51.5 *	51.5	31.5 *	63	51.5 *	51.5	46.5 *	0
Multistage Planning Stage 2	51.5	51.5	20	51.5	51.5	31.5	63	51.5	51.5	63	63
Single-stage Planning	51.5	51.5	20	51.5 *	51.5	31.5 *	63	51.5 *	51.5	63 *	63 *

Important conclusions obtained from the planning include: all the loads were satisfied; at almost all the loads, the voltage requirement was met (5% deviation), except at one long feeder to a heavy load without alternative routes; the network obtained was a radial one with tie-lines normally disconnected; ASAI satisfied the provincial goal of 99.6% and was much better than the present value of 99.1%; new approximate locations for the substations were identified to ease the burden on some existing or planned substations.

Table III Overall Costs and Reliability Indices

	Variable Cost (10 ⁴ ¥) (annual)	Fixed Cost (10 ⁴ ¥) (total)	ASAI
Multistage Planning Stage 1	626.48590	258.24560	99.62014%
Multistage Planning Stage 2	759.35380	1135.72400	99.61013%
Single-stage Planning	756.35360	1265.10100	99.68316%

V. CONCLUSIONS

A new approach that was used to optimize the distribution system of a developing city is described in this paper. Reliability of the system is modelled and optimized in terms of capital costs of outage and reliability investment, along with the costs of the substations and feeders. The nonlinearity iterating method to tackle the nonlinear cost problem and the multi-stage interlacing method to solve the dynamic planning problem are closely interlaced with each other into the network-flow programming procedure. The approach described is practical in solving large-scale problems, while maintaining preciseness by integrating special characteristics of distribution system planning.

For the approach presented in this paper, as well as other systemized distribution system planning methods [1-10], the availability and accuracy of the necessary data are very important in supporting the models. With more and more attention in utilities attracting to the distribution reliability modeling and computerized planning, a more complete and accurate database can be constructed in the near future, through progresses in distribution load forecasting, outage data collecting, operational cost accounting and many other aspects. For utilities that do not currently have the needed data, systemized optimization can still be very valuable in the decision making process, while sensitivity analyses indicating the effects of the input assumptions become necessary.

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References

- [1] D.L.Wall, G.L.Thompson, J.E.D.Northcote-Green, "An Optimization Model for Planning Radial Distribution Networks", IEEE Trans. PAS, Vol. PAS-98, No. 3, May/June 1979, pp. 1061-1068
- [2] T.H.Fawzi, K.F.Ali, S.M.El-Sobki, "A New Planning Model for Distribution Systems", IEEE Trans. PAS, Vol. PAS-102, No. 9, September 1983, pp. 3010-3017
- [3] M.Ponnaivaikko, etc, "Distribution System Planning through a Quadratic Mixed-integer Programming Approach", IEEE Trans. Power Delivery, Vol. PWRD-2, No. 4, October 1987
- [4] Westinghouse Electric Corp., "Research into Load Forecasting and Distribution Planning" (Volume 1), EPRI Project, 1980
- [5] T.Gonen, etc, "Optimal Multi-stage Planning of Power Distribution Systems", IEEE Trans. Power Delivery, Vol. PWRD-2, No. 2, April 1987

[6] I.J.Ramirez-Rosado, T.Gonen, "Pseudodynamic Planning for Expansion of Power Distribution Systems", IEEE Trans. Power Systems, Vol. 6, No. 1, February 1991, pp. 245-254

[7] K.Kara, etc, "Multi-year Expansion Planning for Distribution Systems", IEEE Trans. Power Systems, Vol. 6, No. 3, August 1991, pp. 952-958

[8] R.N.Adams, M.A.Laughton, "A Dynamic Programming / Network Flow Procedure for Distribution System Planning", PICA, 1973, pp. 348-354

[9] S. Karkkainen, etc, "Statistical Distributions of Reliability Indices and Unavailability Costs in Distribution Networks and Their Use in the Planning of Networks", CIRED 1981

[10] G.Kjolle, L.Rolfeseng, E.Dahl, "The Economic Aspect of Reliability in Distribution System Planning", IEEE Trans. Power Delivery, Vol. 5, No. 2, April 1990, pp. 1153-1157

[11] J.L.Kennington, R.V.Helgason, "Algorithms for Network Programming", John Wiley & Sons, 1980

[12] T.Gonen, "Electric Power Distribution System Engineering", McGraw Hill, 1986

[13] R.Billinton, A.N.Allan, "Reliability Evaluation of Power Systems", Pitman Publishing Limited, 1984

[14] J. H. Spare, Minutes of Distribution Reliability Applications Task Force, IEEE PES Winter Meeting, January 31, 1994, New York, NY

Biography

Yifan Tang was born in Fuzhou, China. He received the B.E. and M.E. degrees from Fuzhou University at Fuzhou and Tsinghua University at Beijing, both in electrical engineering, in 1987 and 1990, respectively. He received the Ph.D. degree in electrical engineering from The Ohio State University at Columbus in 1994.

Dr. Tang is currently a Senior Development Engineer with Siemens Power Corporation. His industry experience also includes working as an intern and a consultant for the Shi-Jia-Zhuang Power Company in China from 1988 to 1990, on urban power distribution systems. His teaching and research experience was obtained from 1987 to 1994 at Tsinghua University at Beijing, Northeastern University at Boston and The Ohio State University at Columbus, where he was interested in power distribution system power quality and planning, power electronics, and electric machines, especially with applications of control and systems theory, artificial intelligence and operations research. He is a member of IEEE, and belongs to Power Engineering Society, Industry Application Society and Power Electronics Society.

Discussion

R. Billinton and J. Satish (Power Systems Research Group, University of Saskatchewan): This paper deals with the important problem of reliability optimization in distribution planning. We have a number of questions arising from the paper on which the authors comments would be appreciated.

Equation 2 of the paper incorporates factors such as a - outage cost per MW of outage and b - outage cost per MW of outage per hour of outage. How has the author obtained these parameters?. The material in the paper contains no reference to actual customers which would seem to be of primary importance in distribution planning.

It would be appreciated if you would explain the basis for Equation 22, in which the operation cost is presented as a second order polynomial. The optimization model used in paper minimizes the non linear objective function Z iteratively using a criterion $|Z(k) - Z(k-1)| < \epsilon$. What is the value of ϵ used in this analysis?. It appears that there is a very good possibility of getting sub-optimal and non-optimal solutions with this kind of optimization procedure. It would be appreciated, if the author could indicate how the procedure distinguishes between a local optimum value and the actual global optimum.

In Section 6, the author indicates that voltage drop limits are used as a constraint in the optimization procedure. The method described in the paper uses a network flow model which can not incorporate voltage considerations. It would be appreciated if the author could indicate how the voltage limits were actually incorporated.

The Average Service Availability Index, ASAI is noted in Section 6 as a reliability constraint. The author notes that this index is used "to evaluate the overall reliability level of the optimized network". What value of ASAI is considered by the author as the required limit in the studies described in the paper?

In conclusion we would like to thank the author for his paper and look forward to seeing his response to our questions.

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Y. Tang: The author thanks the discussors for the questions, which will be answered in the order that they were raised.

The outage cost coefficients were the loss cost numbers as seen by and provided by the power company for which part of the work in the paper was done. In obtaining these numbers, as indicated in the paper, individual customer costs and losses were not directly accounted for. The two aspects of outage costs [13,i], i.e., customer losses (direct and indirect) and power company losses (internal and external), were combined into the losses reflected into the more

important (in China) master social-economical balance. In this scheme, the outage costs of different loads were distinguished by the load flows, not by additionally the relative importance of specific users, unless different outage cost coefficients surveyed and categorized across the network were used. With the model and the solution method used in the paper, it is not difficult to incorporate a category of different outage cost coefficients.

The investment costs (equipment, material, engineering, construction, testing, etc.) and maintenance costs of the substations or the feeders are fixed costs dictated by their capacities. The operation costs are second-order functions of their loads, as explained below.

For a feeder, the operation loss costs per kilometer per year can be derived as

$$\begin{aligned}\Delta Z &= C\Delta W = C\Delta P\tau_{\max} \times 10^{-3} \\ &= C 3 (S/1.732/U)^2 R\tau_{\max} \times 10^{-3} \\ &= C (P^2/\cos^2\phi/U^2) R\tau_{\max} \times 10^{-3} \\ &= e_f P^2\end{aligned}$$

where P: load flow (kW);
 ΔP : maximum power losses (kW/km);
 C: value of electricity (Yen/kW/hour);
 U: line voltage level (kV);
 R: feeder resistance (Ω);
 $\cos\phi$: load flow power factor;
 $\tau_{\max}$: feeder branch maximum usage hours;
 e_f : annual operation cost coefficient of feeder branch (element i).

For a substation branch, the operation loss costs per year can be derived as

$$\begin{aligned}\Delta Z &= C\Delta W \\ &= C \tau_{\max} P_k (S^2/S_N^2) \times 10^{-3} \\ &= e_s P^2\end{aligned}$$

where S: load capacity (kVA);
 S_N : substation rated capacity (sum of all main transformers) (kVA);
 P_k : transformer short-circuit power losses (average of all main transformers) (kW);
 $\tau_{\max}$: load maximum usage hours;
 e_s : annual operation cost coefficient of substation branch (element i).

Therefore, the operation costs of the substations or the feeders are second-order functions of their loads, where in (22) e_{ij} represents either e_f or e_s for element i in stage j.

The nonlinear iteration method used in the paper was aiming at practically solving the large-scale optimization problem where computational intensive methods such as nonlinear programming or mixed-integer programming are far from appropriate. With several iterations along with the

multi-stage interlacing, locally the costs $Z_{ij}^{(k)}$ in (23) better approximate the real costs Z_{ij} at the operation points of each branch, than the direct linearization without any modification, while globally the minimum cost was sought in scale by the network-flow optimization and in time by the multi-stage interlacing. Only one global criterion was provided for the nonlinear and multi-stage iterations executed by (28) or (29), and in the reported planning in the paper a value of about 2% required 4 to 5 iterations.

As indicated in the paper, the voltage drop limits were explicitly imposed only after the iterating optimization summarized in the previous paragraph was completed, i.e., the network-flow procedures were not constraint by these limits. All the additional constraints explained in Subsection 6) of Section III were checked after the optimization, and in case that major modifications were required, a new stage-by-stage simple network-flow optimization might be necessary (see the Flow-chart), or even a totally new re-run of the main program in extreme cases after modifying some input data.

Similar answer can be given regarding the reliability constraint ASAI, which in the planning was the provincial goal of 99.6% (satisfied by the planning without system re-adjustments). The calculations of the reliability indices (annual outage rates, average outage duration and annual outage duration) for the loads and the ASAI should consider

the effects of switches, circuit breakers and tie-line load transfer, and the details were given in [13,i].

Given the dimension of the work and the difficulty of the task, a single paper such as this would not be able to cover all the aspects sufficiently other than suggesting a framework for further developments. More derivations and philosophy, though far from solving the difficult problem of systemized optimization, had been outlined in [ii]. Finally, the author would like to take this opportunity to correct two typos in the paper. One in Equation (22), where the first summation should be from " $j=1$ ", as in (12). Another one in Fig. 5(b), where the term " P_1-P_L " should read " P_L-P_1 ". Any inconvenience is apologized.

References for Closure

- [i] Guo Y., "Power System Reliability: Principle and Application" (in Chinese), Vol. I and II, Tsinghua University Press, Beijing, China, 1986
- [ii] Y. Tang, "Study of Power Distribution System Planning" (in Chinese), M.Eng. Thesis, Tsinghua University, Beijing, China, March 1990

Manuscript received April 17, 1995.